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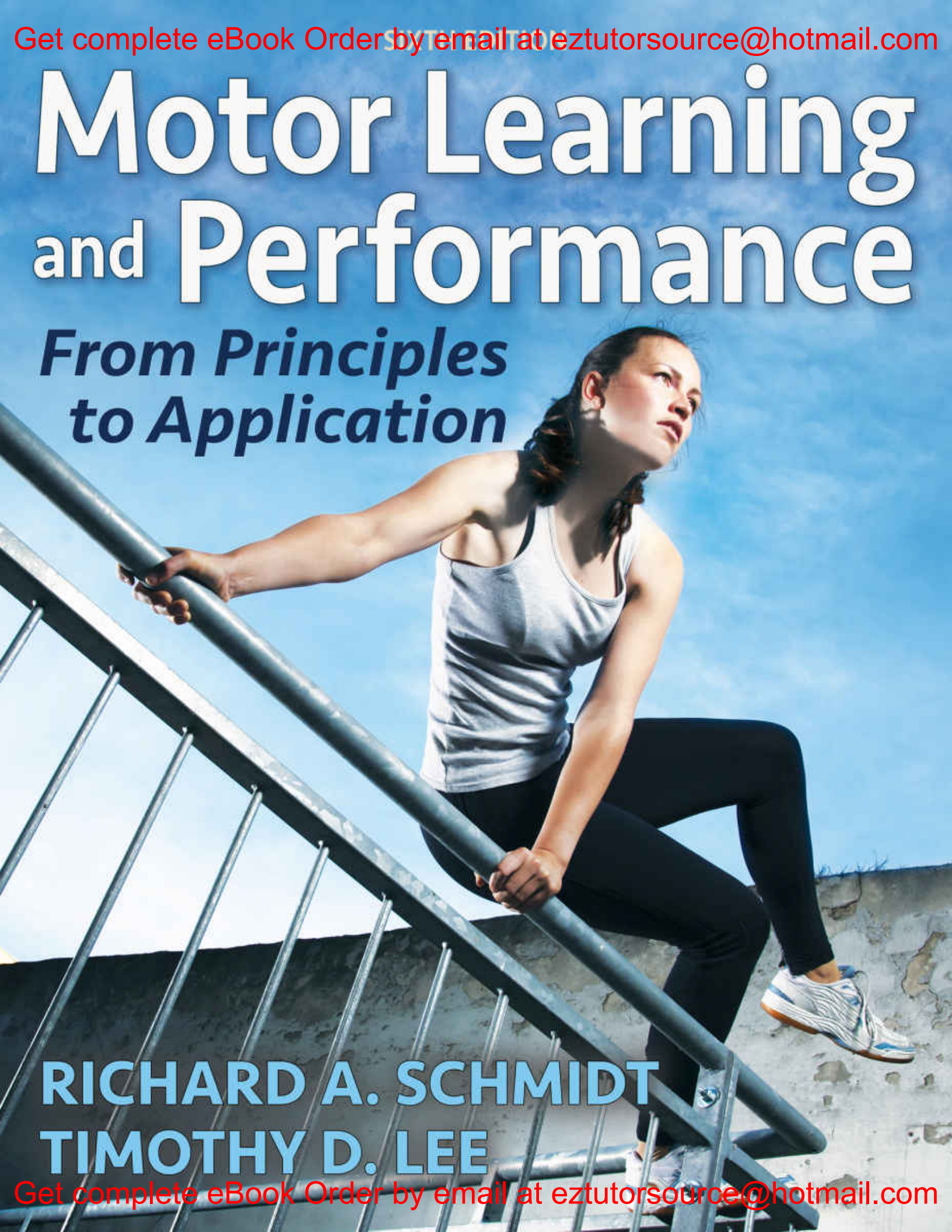
From Principles to Application



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From Principles to Application

SIXTH EDITION

Richard A. Schmidt
Timothy D. Lee

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1

Introduction to Motor Learning and Performance

How Skills Are Studied

KEY TERMS
absolute constant error (|CE|)
absolute error (AE)
closed skill
constant error (CE)

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continuous skill
correlation coefficient
discrete skill
open skill
root-mean-square error (RMSE)
serial skill
skill
tracking
variable error (VE)

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CHAPTER OUTLINE

Why Study Motor Skills?
The Science of Motor Learning and Performance
Defining Skills
Components of Skills
Classifying Skills
Measuring Skilled Performance
Understanding Performance and Learning
Summary

CHAPTER OBJECTIVES

Chapter 1 provides an overview of research in human motor skills with particular reference to their study in motor learning and performance. This chapter will help you to understand

- the scientific method in skills research,
- different taxonomies used to classify skills,
- common variables used to measure motor performance, and
- the rationale for developing a conceptual model of motor performance.

Parkour videos can be scary to watch, let alone being a performer in one. The traceur or traceuse performs acts of balance, agility, and dexterity, usually across great expanses of space and often at dizzying heights above ground. And yet they perform these skills with grace and confidence. How people perform these skills at such high levels, how such skills are developed, and how you can develop some approximation of these skills in yourself, your children, or your students—all of these questions generate fascination, encouraging further understanding about human movement.

A description of the study of motor performance and learning starts here. The overview presented in this chapter introduces the concept of skill and discusses various features of its definition. The chapter then gives examples of skill classification schemes important for later applications. Finally, the logic behind the book's organization is

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described to help you understand skills effectively: first, the principles and processes underlying skilled performance, followed by how such capabilities can be developed with practice.

The remarkable human capability to perform skills is a critical feature of our very existence. It is almost uniquely human, although various animals relatively high on the evolutionary scale can be trained to produce what you might call skilled behaviors (e.g., circus dogs doing somersaults, bears riding bicycles). Without the capacity for skilled performance, we could not type the page we are preparing now and you could not read it. And for students involved in physical education and kinesiology, coaching, physical (or speech or occupational) therapy, chiropractic, medicine, or human factors and ergonomics, here is the opportunity to learn about the fundamentals of a wide variety of sports and athletic endeavors, music, and simply ordinary everyday actions that are so strongly fascinating and exciting.

Human skills take many forms, of course—from those that emphasize the control and coordination of our largest muscle groups in relatively forceful activities such as soccer or tumbling, to those in which the smallest muscle groups must be tuned precisely, as in typing or repairing a watch. This text generally focuses on the full range of skilled behavior because it is useful to understand that many common features underlie the performance of skills associated with industrial and military settings, sport, the reacquisition of movement capabilities lost through injuries or stroke, or simply the everyday activities of most people.

Most humans are born with the capability to produce many skills, and only some maturation and experience are necessary in order to produce them in nearly complete form. Walking and running, chewing, balancing, and avoiding painful stimuli are some examples of these relatively innate behaviors. But imagine what simple and uninteresting creatures we would be if these inherited actions were all that we could ever do. All biological organisms have the remarkable facility to profit from their experiences, to learn to detect important

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environmental features (and to ignore others), and to produce behaviors that were not a part of their original capabilities. Humans seem to have the most flexibility of all, which allows gains in proficiency for occupations as chemists or computer programmers, for competition in music or athletics, or simply for conducting daily lives more efficiently. Thus, producing skilled behaviors and the learning that leads to their development are tightly intertwined in human experience. This book is about both of these aspects of skills—skilled human performance and human learning.

This book is *not* about skills in which the degree of success is determined by deciding which of many already learned actions the performer is to do. When the laboratory rat learns to press a bar at the presentation of a sound, the rat is not learning how to press the bar; rather, the animal is learning when to make this already learned bar-pressing action. As another example, in the card game of bridge, it does not matter *in what fashion* the various cards are played (i.e., moved). What matters is cognitive decision making about which card to play and when to do it. The study of these kinds of decision-making processes falls mainly into fields such as experimental psychology and cognitive neuroscience, and these processes are deliberately not included here.

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Why Study Motor Skills?

Because skills make up such a large part of human life, scientists and educators have been trying for centuries to understand the determinants of skills and the factors that affect their performance. The knowledge gained is applicable to numerous aspects of life. Important points apply to the instruction of skills, where methods for efficient teaching and effective carryover to life situations are primary concerns. There is also considerable applicability for improving high-level performances, such as sport, music, and surgical skills. Of course, much of what coaches and music and physical education teachers do during their professional activities involves, in one way or another, skills instruction. The practitioners who understand these skill-related processes most effectively undoubtedly have an advantage when their trainees perform their activities.

Other application areas can be emphasized as well. There are many applications in training skills for industry, where effective job skills can mean success in the workplace and can be major determinants of satisfaction both with the job and with life in general. Teaching job skills most effectively, and determining which of a large number of individuals are best suited to particular occupations, are common situations in which knowledge about skills can be useful in industry. Usually these applications are considered within human factors (ergonomics). The principles also apply to physical therapy and occupational therapy settings, where the concern is for the (re)learning and production of movements that have been lost through head or spinal cord injury, stroke, birth defects, and the like. Although all these areas may be different and the physical capabilities of the learners may vary widely, the principles that lead to successful application are generally the same.

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The Science of Motor Learning and Performance

It is not uncommon that as an area of interest grows, the systematic study of the principles involved also develops. Motor learning and performance are no different, in that a science has emerged that allows the formalization of terms and concepts for others to use. When we use the word *science*, what do we mean?

The concept of a science implies several things: (1) the active use of theory and hypothesis testing to further our knowledge, (2) a certain infrastructure that involves books and journals, scientific organizations that deal with both the fundamental aspects of the science and ways to apply the knowledge to real-world situations, and granting agencies to provide funds for research, and, of course, (3) the existence of courses of study of the area in universities and colleges.

Theories and Hypotheses

Certainly at the heart of every science are theories that purport to explain how things work. A theory's purpose is to explain how phenomena occur. The theorist devises *hypothetical constructs*—elements or pieces that interact in various ways in the theory. The theorist then describes the ways in which the hypothetical constructs interact with each other so as to explain some empirical phenomenon. Then, using logical deduction, scientists determine certain predictions that the theory makes in its current form. These predictions form the basis of hypotheses that can be tested, typically in the laboratory. These hypotheses take the form of statements such as "If I ask learners to practice under condition *x*, then learning should be better than for others who practice under condition *y*."

Theories are typically tested by doing experiments, which determine whether or not the hypothesis correctly predicted what happened. In the field of motor learning and performance, these experiments typically take the form of having at least two groups of research participants randomly assigned to experimental treatments,

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with one group performing a task under condition *x* (as in the example just mentioned) and the other group performing under some other conditions that are reasonably well understood (sometimes called a control condition). If, in this example, the group practicing under condition *x* outperforms the group in the control condition, then we say that the hypothesis is supported. However, a given theory might predict an outcome that would make sense for several different theories, so an experiment that supports the hypothesis cannot be said to *prove* that is theory is correct. What is usually far stronger is if the results come out *contrary* to the prediction; this leads to the logical inference that the theory must be incorrect, allowing us to reject one of the possible theories. This is so because a theory cannot survive for long if something predicted from it turns out not to be the case. Because of this difference in the power of the ways in which hypotheses are tested, scientists tend to search out predictions from a theory that might not hold if tested in the laboratory.

What Skills Have Been Studied and Who Studies Them?

The science of motor learning and performance has been used to study many varieties of skills. Initially, skills research was primarily the domain of psychologists and physiologists who used movement tasks as tools to examine theories of learning and coordination. For example, in the 1890s, Bryan and Harter studied the complex processes involved in sending and receiving Morse code. In the 1920s, the psychologist E.L. Thorndike used blindfolded line drawing to examine the role of augmented feedback to test his Law of Effect. Rotary pursuit tracking was a common tool used to examine Hull's theory of learning in the 1940s. And simple line-drawing tasks came back into popularity again in the 1960s to study the forgetting of information over short retention periods.

Focus on
RESEARCH 1.1

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Paul M. Fitts

Have you ever wondered why some keys on a calculator or your computer keyboard are larger than others, or why they are arranged to occur in specific locations? If so, then you can probably credit Paul Fitts with the underlying rationale. His research was designed to answer fundamental questions about human performance in situations that required speed and accuracy, often under stressful situations. Some of his many contributions included research on stimulus–response compatibility effects (see chapter 2), the effects of target size and distance on movement time, now known as Fitts' Law (chapter 6), and the stages of learning (chapter 9). His work was largely responsible for the growth of the research area now called ergonomics or human factors.

In the late 19th century, physiologists such as Sir Charles Sherrington were interested in the role of various biological systems concerning the fundamental mechanisms of neural control of muscle and the study of the nervous system. Nikolai Bernstein and Erich von Holst contributed essential research to our understanding of movement coordination during the first half of the 20th century. And Edward Taub performed controversial experiments on monkeys that revealed evidence about lost and retained movement capabilities in the absence of sensory feedback.

Perhaps motivated by discoveries in these more traditional fields, researchers in other disciplines began to study motor skills as a means to answer domain-specific issues. Paul Fitts (see Focus on Research 1.1) contributed to the war effort in the 1940s by studying piloting errors—why errors occurred and how movements are limited in speed and accuracy by task demands. Fitts' work represented an emerging field called ergonomics (also called human factors), which

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broadly studied the limits of human performance in the execution of industrial tasks.

Physical education and kinesiology departments, led by researchers such as Franklin M. Henry (see Focus on Research 1.2), studied motor skills as a research topic of direct interest in and of itself. Later, researchers added to this effort by establishing testable theories regarding the learning and control of motor skills.

It is common now to find research being conducted in various fields of application in which skills are used as components of daily living, such as medicine, dentistry, physical rehabilitation, fire prevention, law enforcement, and the interface of humans with various forms of computing devices. And, of course, the study of how highly skilled performance is acquired and executed remains an ongoing field of study in sport and music. All of these influences, in the pursuit of both practical and theoretical advances to our knowledge, have served to drive the study of motor learning and performance.

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Defining Skills

As widely represented and diverse as skills are, it is difficult to define them in a way that applies to all cases. Guthrie (1952) provided a definition that captures most of the critical features of skills that we emphasize here. He defined **skill** as "the ability to bring about some end result with maximum certainty and minimum outlay of energy, or of time and energy" (p. 136). Next, we consider some of the important components (or features) of this definition.

First, performing skills implies some desired goal, such as holding a handstand in gymnastics, expressing thoughts using a keyboard, or being able to walk again after a stroke. Skills are usually thought of as different from *movements*, which do not necessarily have any particular goal, such as idly wiggling one's little finger. Of course, skills consist of movements because the performer could not achieve the goal without making at least one movement.

Focus on
RESEARCH 1.2

Franklin M. Henry, Father of Motor Behavior Research

Before World War II and during the 1950s and 1960s when much effort was directed at military skills such as pilotry, most of the research in movement behavior and learning was done by experimental psychologists, studying relatively fine motor skills. Little effort was devoted to the gross motor skills that would be applicable to many sports. Franklin M. Henry, trained in experimental psychology but working in the Department of Physical Education at the University of California, Berkeley, was filling this gap with a new tradition of laboratory experimentation that started an important new

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direction in research on movement skills. He studied gross motor skills, with performances intentionally representative of those seen on the playing fields and in gymnasiums. But he used laboratory tasks, which enabled the rigorous study of these skills employing methods analogous to those used in experimental psychology. Two of his major contributions to research were (1) evidence that initiation times increase as movements-to-be-made increase in complexity, providing evidence in support of the motor program concept (see chapter 5), and (2) evidence that skills are highly uncorrelated within individuals (see the section "Correlation: The Association Strength Between Scores" later in this chapter), which dispelled the general motor ability concept (see chapter 7).

Second, to be skilled implies meeting this performance goal, this end result, with maximum *certainty*. For example, suppose that a darts player makes a bull's-eye. This by itself does not ensure that he is a skilled darts player, because there is no evidence that this result was achieved with much certainty. Such an outcome could have been the result of one lucky throw in the midst of hundreds of others that were not so lucky. To be considered skilled requires that a person produce the skill reliably, on demand, without luck playing a very large role. This is one reason why people value so greatly the champion athlete who, with but one chance and only seconds remaining at the end of a game, makes the play that allows the team to win.

Third, a major feature in many skills is the minimization, and thus conservation, of the energy required for performance. For some skills this is clearly not the goal, as in the shot put, where the only goal is to achieve the maximum distance. But for many other skills the minimization of energy expenditure is critical, allowing the marathon runner to hold an efficient pace or allowing the wrestler to save strength for the last few minutes of the match. We evolved to walk as

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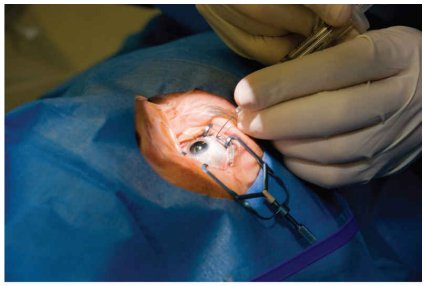
we do, in part, because our walking style minimizes energy expenditure for walking a given distance. This minimum-energy notion applies not only to the physiological energy costs but also to the psychological or mental energy required for performance. Many skills have been learned so well that the performers hardly have to pay attention to them, freeing their cognitive processes for other features of the activity, such as strategic planning in team sports or artistic expressiveness in dance and music. A major contributor to the efficiency of skilled performance is, of course, practice, with learning and experience leading to the relatively effortless performances so admired in highly skilled people.



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Identify the main emphases in each of these photos with respect to the definition of skill.

Finally, another feature of many skills is for highly proficient performers to achieve their goals in minimum time. Many skills have this as the only goal, such as a swimming race. Minimizing time can interact with the other skill features mentioned, however. Surgeons who conduct invasive surgery need to work quickly to minimize the opportunity for infections to enter the body. Yet surgeons obviously need to work carefully too. Speeding up performance often results in imprecise movements that have less certainty in terms of achieving their goals. Also, increased speed generates movements for which the energy costs are sometimes higher. Thus, understanding skills involves optimizing and balancing several skill aspects that are important to different extents in different settings. In sum, skills generally involve achieving some well-defined goal by

- maximizing the certainty of goal achievement,
- minimizing the physical- and mental-energy costs of performance, and
- minimizing the time used.

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Components of Skills

The elegant performance of the skilled dancer and the artistic talents of an expert sculptor may appear simple, but the performance goals actually were realized through a complex combination of interacting mental and motor processes. For example, many skills involve considerable emphasis on sensory–perceptual factors, such as detecting that a tennis opponent is going to hit a shot to your left or that you are rapidly approaching a car that has suddenly stopped on the road ahead of you. Often, sensory factors require the split-second analysis of *patterns* of sensory input, such as discerning that the combined movements of an entire football team indicate that the play will be a running play to the left side. These perceptual events lead to decisions about what to do, how to do it, and when to do it. These decisions are often a major determinant of success. Finally, of course, skills typically depend on the quality of movement generated as a result of these decisions. Even if the situation is correctly perceived and the response decisions are appropriate, the performer will not be effective in meeting the environmental goal if she executes the actions poorly.

These three elements are critical to almost any skill:

- Perceiving the relevant environmental features
- Deciding what to do and where and when to do it to achieve the goal
- Producing organized muscular activity to generate movements that achieve the goal

The movements have several recognizable parts. Postural components support the actions; for instance, the arms and hands of a surgeon need to be supported by a stable platform in order to perform accurately. Body transport components move the body toward the point where the skill will take place, as in carrying a package of shingles up a ladder to place on the roof of a house.

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It is interesting, but perhaps unfortunate, that each of these skill components seems to be recognized and studied in isolation from the others. For example, sensory factors in perception are studied by cognitive psychologists, scientists interested in (among other things) the complex information-processing activities involved in seeing, hearing, and feeling. Sometimes these factors are in the realm of psychophysics, the branch of psychology that examines the relationship between objective physical stimuli (e.g., vibration intensity) and the subjective sensations these stimuli create when perceived (e.g., loudness). Factors in the control of the movement itself are typically studied by scientists in the neurosciences, cognitive psychology, biomechanics, and physiology. Skill learning is studied by yet another group of scientists in kinesiology, in experimental or educational psychology, or in the field of human factors and ergonomics. A major problem for the study of skills, therefore, is the fact that the several components of skill are studied by widely different groups of scientists, often with little overlap and communication among them.

All of these various processes are present in almost all motor skills. Even so, we should not get the idea that all skills are fundamentally the same. In fact, the principles of human performance and learning depend to some extent on the *kind* of movement skill to be performed. So the ways in which skills have been classified are discussed next.

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Classifying Skills

There are several skill classification systems that help organize the research findings and make application somewhat more straightforward. These are presented in the following sections.

Open and Closed Skills

One way to classify movement skills concerns the extent to which the environment is stable and predictable throughout performance. An **open skill** is one for which the environment is variable and unpredictable during the action. Examples include most team sports and driving a car in traffic where it is difficult to predict the future actions of other people. A **closed skill**, on the other hand, is one for which the environment is stable and predictable. Examples include swimming in an empty lane in a pool and drilling a hole in a block of wood. These open and closed designations actually mark only the end points of a continuum, with the skills lying in between having varying degrees of environmental predictability or variability (see Gentile, 2000, for more discussion).

This classification points out a critical feature for skills, defining the performer's need to respond to moment-to-moment variations in the environment. It thus brings in the subprocesses associated with perception, pattern recognition, and decision making (usually with the need to perform these processes quickly) so the action can be tailored to the environment. These processes are minimized in closed skills, where the performer can evaluate the environmental demands in advance without time pressure, organize the movement in advance, and carry it out without needing to make rapid modifications as the movement unfolds. These features are summarized in table 1.1.

TABLE 1.1 Open and Closed Skills Continuum

Closed skills		Open skills
Predictable environment	Semipredictable environment	Unpredictable environment

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Gymnastics	Walking a tightrope	Playing soccer
Archery	Steering a car	Wrestling
Typing	Playing chess	Chasing a rabbit

Discrete, Serial, and Continuous Skills

A second scheme for classifying skills concerns the extent to which the movement is an ongoing stream of behavior, as opposed to a brief, well-defined action. At one end of this dimension is a **discrete skill**, which usually has an easily defined beginning and end, often with a very brief duration of movement, such as throwing a ball, firing a rifle, or turning on a light switch. Discrete skills are particularly important in both sport and daily actions, especially considering the large number of discrete hitting, kicking, and throwing skills that make up many sport activities, as well as everyday skills of fastening buttons, writing your signature, and pointing a computer mouse. Discrete skills often result in a measured outcome score, which can be combined with other scores to result in several types of error scores, discussed later in this chapter.

At the other end of this dimension is a **continuous skill**, which has no particular beginning or end, the behavior flowing on for many minutes, such as swimming or knitting. As discussed later, discrete and continuous skills can be quite different, requiring different processes for performance and demanding that they be taught somewhat differently as a result.

One particularly important continuous skill is **tracking**, in which the performer's limb movements control a lever, a wheel, a handle, or some other device to follow the movements of a target. Steering a car involves tracking, with steering wheel movements made so the car follows the track, defined by the roadway. Tracking movements are very common in real-world skills situations, and much research has been directed to their performance and learning. Tracking tasks are sometimes scored using a particular error score, called root-mean-square error (RMSE) presented in detail later in this chapter.

Between the polar ends of the discrete–continuous skill continuum is the **serial skill**, which is often thought of as a group of discrete

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skills strung together to make up a new, more complicated skilled action. See table 1.2 for a comparison summary. Here the word *serial* implies that the order of the elements is usually critical for successful performance. Cars that have a manual transmission involve a serial skill that combines discrete arm movements along with movements of both feet performed in a precisely timed sequence to create a larger action. Other examples include performing a gymnastics routine and most types of cooking. Serial skills differ from discrete skills in that the movement durations tend to be somewhat longer, yet each component retains a discrete beginning and end. One view of learning serial skills suggests that the individual skill elements present in early learning are somehow combined to form one larger, single element that the performer controls almost as if it were truly discrete in nature (e.g., the smooth, rapid way a gymnast shifts from one maneuver to another on the rings).

TABLE 1.2 Discrete–Serial–Continuous Skills Continuum

Discrete skills	Serial skills	Continuous skills
Distinct beginning and end	**Discrete actions linked together**	**No discrete beginning or end**
Throwing a dart	Hammering a nail	Steering a cart
Flipping a light switch	Assembly-line work	Swimming
Shooting a rifle	Gymnastics routine	Tracking task

Object Manipulation and Body Transport

Gentile (2000) identified two other factors that help to more precisely define the nature of skills. One concerns whether the body is stationary or in motion during the performance of the skill. For example, a dentist usually performs her work while standing still, in contrast to a table busser in a restaurant, who is constantly on the move. The other factor is whether or not the performer is manipulating an object during performance. Figure skaters express artistry in motion on ice skates, in contrast to puck control, which is the primary goal in ice hockey.

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The four factors described above define skills as a complex classification system. Ultimately, almost every skill that you can think of can be described as an interacting product of these four dimensions.

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Measuring Skilled Performance

Quite frequently, researchers are called on to generate methods for computing scores for a participant who was attempting to perform a series of trials on a test. As we shall see, there are various ways to calculate these measures, depending on whether the task is discrete or continuous.

Error Scores in Discrete Tasks

Suppose that you are testing individuals on a throwing task, in which they are trying to throw a ball exactly 50 ft away from where they are currently standing. Two hypothetical participants perform this task for five trials, and the results are illustrated in figure 1.1 and presented in table 1.3.

Which of these two individuals was more skillful in this task? It appears from the figure that there is a difference in skill, but looking at a figure and making decisions based on one's impression is less scientific and precise than developing measures that are based on the actual numbers and inferential statistics. Therefore, researchers have come up with ways of combining the scores on the five throwing trials into a single number that reflects their skill in the task. Several candidate measures are possible, and each represents a unique assessment of skill.

The worksheet presented in table 1.3 presents a detailed description of how to calculate various methods that represent the skills illustrated by the two throwers in figure 1.1.

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Constant Error (CE)

The simplest way to determine which person was more accurate is to compute the error deviation of each throw, relative to the target, and then calculate the average of these error deviations (column 4 in table 1.3). For example, on trial 1, Chester's error score was −4 (his

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throw was 46 ft, and thus, 4 ft short of the target of 50 ft). The error score on the second trial was +2 (his 52 ft throw was 2 ft too far), and so on. After the error scores for each trial are computed, the mean can then be calculated to determine the average error deviation from the target. This is termed the participant's average **constant error (CE)**. The interpretation is that Chester tended to overthrow the 50 ft target by only 0.4 ft; John-Lee tended to underthrow the target by 2.6 ft, therefore Chester tended to be closer to the goal than John-Lee, based on average CE.

Classify the actions depicted here along the four skill classification dimensions: discrete/continuous/serial; open/closed; body in motion or stationary; object or no object.

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The formula for constant error is

$$CE = [\Sigma\ (X_i - T)\ /\ N]$$

where: Σ = the "sum of," i = trial number, X_i = score for the ith trial, T = the target distance, and N = the number of trials.

Absolute Error (AE)

Another simple way to combine the scores into a single number is to consider the *absolute value* (i.e., with the sign ignored or removed) of the error on each trial, and take the average of those unsigned error scores for the various trials (column 5 in table 1.3). For example, Chester has an error of −4 ft on the first trial; when we take the absolute value, the first trial has an absolute error of 4 ft; the second trial has an error of +2 ft, whose absolute value is 2. If we do this procedure for the remainder of the trials for Chester, and all the trials for John-Lee, the computed average absolute error for Chester is 6.4; for John-Lee it is 4.2. Here, with the direction of the errors being disregarded, the interpretation is that Chester was off-target more than was John-Lee.

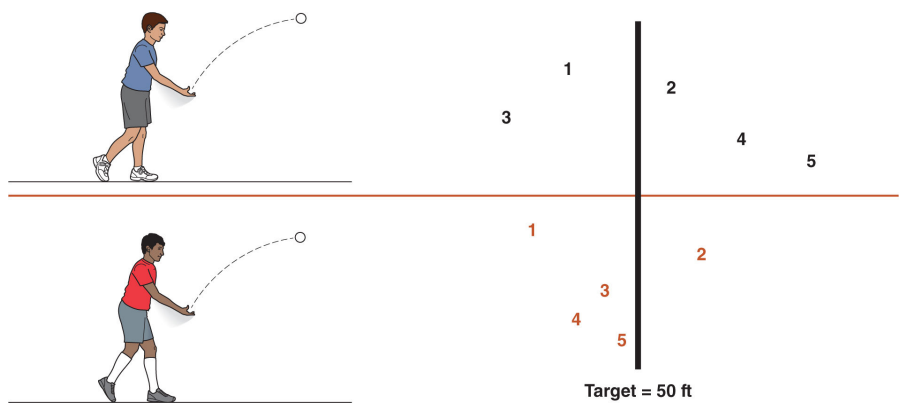

FIGURE 1.1 Two performers are attempting to throw a ball exactly 50 feet. The numbers (1–5) represent the individual results of each of five trials. The numerical results are presented in table 1.3.

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TABLE 1.3 Error Scores in Discrete Tasks

Chester					
			CE	AE	VE
Trial	X_i	T	$X_i - T$	$\lvert X_i - T \rvert$	$(X_i - CE)^2$
1	46	50	−4	4	19.36
2	52	50	+2	2	2.56
3	39	50	−11	11	129.96
4	55	50	+5	5	21.16
5	60	50	+10	10	92.16
			$\Sigma(X_i - T) = 2.0$	$\Sigma(\lvert X_i - T \rvert) = 32.0$	$\Sigma(X_i - CE)^2 = 265.2$
			$\Sigma(X_i - T)/N = +0.4$	$\Sigma(\lvert X_i - T \rvert)/N = 6.4$	$\Sigma(X_i - CE)^2 / N = 53.04$
					$\sqrt{[\Sigma(X_i - CE)^2 / N]} = 7.3$
John-Lee					
			CE	AE	VE
Trial	X_i	T	$X_i - T$	$\lvert X_i - T \rvert$	$(X_i - CE)^2$
1	40	50	−10	10	54.76
2	54	50	+4	4	43.56
3	48	50	−2	2	0.36
4	46	50	−4	4	1.96
5	49	50	−1	1	2.56
			$\Sigma(X_i - T) = -13$	$\Sigma(\lvert X_i - T \rvert) = 21$	$\Sigma(X_i - CE)^2 = 103.2$
			$\Sigma(X_i - T)/N = -2.6$	$\Sigma(\lvert X_i - T \rvert)/N = 4.2$	$\Sigma(X_i - CE)^2 / N = 20.64$
					$\sqrt{[\Sigma(X_i - CE)^2 / N]} = 4.5$

The formula for average **absolute error (AE)** is

$$AE = [\Sigma\,(|X_i - T|) / N]$$

where: Σ, *i*, *X*, *T*, and *N* are defined as before for constant error, and the vertical bars (|) mean "absolute value of."

Variable Error (VE)

The third measure of skill is not actually a measure of accuracy but rather an assessment of the participant's inconsistency—that is, how different each individual score was in comparison to his average (CE) score. To compute **variable error (VE)**, we square the difference between each trial's error score and the individual's own constant error $[(X_i - CE)^2]$, sum those squared values over all of the trials, and divide by *N*. Now, since these are squared values, we return them to

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their original state by computing the square root of this value (see column 6 in table 1.3).

In this example, for Chester's first trial, the difference between X_1 and Chester's average CE (which was +0.4) is [(−4 − (+0.4)] = −4.4 ft, which, when squared, is 19.36 ft. For the second trial, the difference between X_2 and Chester's average CE is [(+2) − (+0.4)] = 1.6 ft, which, when squared, is 2.56 ft. Now do the same thing for trials 3, 4, and 5; add up all five squared differences; divide this number by N = 5; then take the square root of that number; and you finally have VE. The score is interpreted as the inconsistency in responding—that is, how variable the performance is about the person's own average CE. Of course when computing the value for John-Lee, we would use his CE (which was −2.6) in the computations. The computed VE score for Chester was 7.3 ft, and for John-Lee the VE was 4.5 ft. The interpretation is that even though Chester's average throw was closer to the goal than was John-Lee's average throw, Chester was more inconsistent in those throws than was John-Lee.

The formula for variable error is

$$VE = \sqrt{[\Sigma\,(X_i - CE)^2 / N]}$$

where Σ, *X, i, CE,* and *N* are defined as before, and √ is the square root.

AE was employed as a measure of error very frequently until investigators Schutz and Roy (1973) pointed out some statistical difficulties with it. Today, investigators tend to use CE (as a measure of average bias or directional error) and VE (as a measure of inconsistency). Sometimes investigators use a statistic called **absolute constant error** (abbreviated **|CE|**) instead of CE for an individual's measure of bias. The |CE| is simply the average absolute value of the computed CE score as defined previously. The advantage is that |CE| retains the magnitude of average deviation from the target but prevents two scores from canceling out when data from more than one person are averaged together to present a group

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score. Much more on these error scores is presented in chapter 2 of Schmidt et al. (2019).

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Error Scores in Continuous Tasks

Continuous tasks, such as tracking, are capable of producing many error scores on a single trial. Consider figure 1.2 as an example of a portion of a single trial of a tracking task for one individual. The wide blue path represents the goal, such as a highway lane along which a driver might navigate a car. The red line represents the exact (but usually unmarked) center of the track from edge to edge. The black dotted line represents a person's tracking behavior—in this case, how close to the center of the road the car is maintained. How can skill be measured in this single trial of a continuous performance?

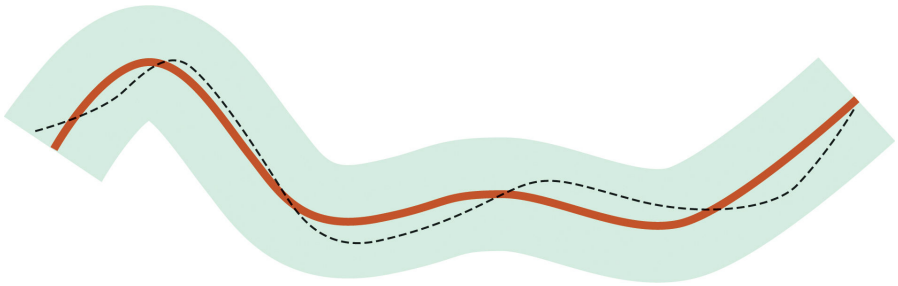

FIGURE 1.2 Measuring RMSE in a tracking task. The blue area represents the track (such as a highway lane); the red (solid) line represents the unmarked center of the lane, and can be considered the participant's goal track. The black dotted line represents the participant's performance in attempting to follow the red line.

A common method used by researchers who study tracking tasks is to compute a measure called **root-mean-square error (RMSE)**. One does this by computing the distance of the person's tracking behavior from the target line at preset *distance points* along the track (e.g., every 10 ft of highway traveled) or, more commonly, at a constant interval of time along the track (e.g., every 100 ms). This method effectively slices the continuous movement into equal, discrete intervals of tracking behavior, from start to finish.

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With each slice of the track, the researcher then computes how far the participant's tracking position is from the target. Since the red line in figure 1.2 represents the center of the track, it is convenient to define the red line as the zero position. Therefore, if the participant is to the right of the target, the measure is given a positive error value; if the tracking position is to the left of the target, the measure gets a negative value. The root-mean-square error score is computed by first calculating the squared deviations for each measured position along the track, then taking the square root of the sum of those scores.

The RMSE is a more complex measure of performance than any of the error scores for discrete tasks, because it represents two components of behavior. The RMSE reflects both the bias tendency (e.g., on average, to drive closer to the right edge of the lane than the left edge) as well as inconsistency in the tracking behavior (how variable the performance tends to be). The RMSE is well recognized as a very good measure of how effectively the person tracked.

Correlation: The Association Strength Between Scores

Sometimes researchers are interested in discovering how sets of scores are related to each other. Franklin M. Henry (see Focus on Research 1.2) was an early pioneer of using correlations as a measure of determining one such relationship. His reasoning was this: Assume that a relatively large number of people are tested on each of two skills, A and B. Henry reasoned that if one person was an outstanding performer on skill A, then this person would be assumed to have a strong general motor ability. If so, this person should also score well on skill B, which also depended on the general motor ability. Conversely, if another person did not score well on skill A, at least part of the reason would be that this person had a weak general motor ability, and this person would be expected to score relatively poorly on skill B also. In this way, skill A and skill B are *related* to each other, in that good scores on A go with good scores on B, and poor scores on A go with poor scores on B.

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With this kind of relationship, if we were to plot performances of skill A against performances of skill B, as we have done in figure 1.3, where each dot represents a single individual measured on both tests A and B, these two tests should plot linearly with each other, which they tend to do in figure 1.3*a*. However, if skill A and skill B are not related to each other, then they should plot more or less as seen in figure 1.3*b*. One interpretation of these scatter plots is that in figure 1.3*a*, skill A and skill B tend to be measures of the same thing, and the usual interpretation is that they are both measures of the same ability. In figure 1.3*b*, on the other hand, we would be forced to say that skill A and skill B are not measures of the same ability. This leads to the straightforward prediction that if general motor ability exists, with all skills being dependent on a single general motor ability, then all skills should show strong relationships among them, as in figure 1.3*a*.

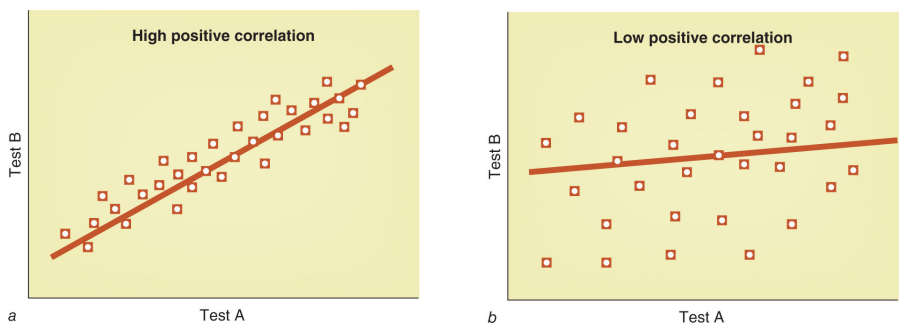

FIGURE 1.3 Scatter plot of two tests revealing a high positive correlation (*a*) and a very low positive correlation (*b*). Each dot in the figure represents the performance of one individual, plotting performance on one test against performance on the other test.

As we mentioned previously, however, determining conclusions based on the visual inspection of trends in data is imprecise. Fortunately, a statistic called the **correlation coefficient** provides a way to determine the exact strength of a relationship, for example, between the data that are illustrated in figure 1.3.

The correlation can range in size from −1.0 to +1.0. The size of the correlation indicates the strength of the relationship, or how close the individual dots are to the best-fitting line passing through them. If the line is sloped negatively—downward to the right—then the sign of the

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correlation will be negative. If the dots are close to the line, as they are in figure 1.3*a*, the correlation is close to +1.0, indicating a very strong tendency for skill in A to be associated with skill in B (r = +.90). If the dots lie relatively far from the line, as they do in figure 1.3*b*, the correlation is closer to zero, indicating a relatively weak tendency for scores in A to be associated with scores in B (r = +.15). The strength of a relationship is estimated by the squared correlation coefficient multiplied by 100. Thus, the correlation of +.15 means that the two tests have $.15^2 = (.15 \times .15) \times 100$, or about 2% in common with each other. Note that the size of the correlation has nothing to do with its sign, because a strong correlation can be either positive or negative. Finally, a correlation of .00 indicates that the line of best fit is a horizontal line (a slope of 0), with the dots scattered about it in a random way. In such a case, tests A and B would not be related at all. We will have much more to say about the findings uncovered by this research in chapter 7.

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Understanding Performance and Learning

In some ways, skilled performance and motor learning are interrelated concepts that cannot be easily separated for analysis. Even so, a separation of these areas is necessary for presentation and making eventual understanding much easier. Many of the terms, principles, and processes that scientists use to describe the improvements with practice and learning (the subfield of motor learning) actually come from the literature on the underlying processes in the production of skilled motor performance (the subfield of human performance; the field of motor behavior often includes both motor learning and human performance). Therefore, whereas at first glance it might seem most logical to treat motor learning before motor performance (a person has to learn before performing), it turns out to be awkward to present information on learning without first having provided this background performance information.

For this reason this book is organized into two parts, the first of which introduces the terminology, concepts, and principles related to skilled human performance without very much reference to processes associated with learning. The principles here probably apply most strongly to the performance of already skilled actions. Having examined the principles of how the motor system produces skills, the discussion turns to how these processes can be altered, facilitated, and trained through practice, which involves motor learning.

Figure 1.4 illustrates a simplified relationship between motor learning and performance in the context of the issues discussed in this chapter. Measurable performance is the result of processes that culminate in movement. Just as the skill has different identifiable dimensions, so too does the individual, and the first part of this book is devoted to describing the processes that influence motor performance. Learning, which represents a relatively permanent improvement in the performance of a skill, occurs when those processes are repeated and refined, which is the topic of the second part of this book.

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When studying motor performance and learning, it is helpful to understand where each concept fits into the complex process of performing a skill. For this reason, and to help apply skills information to a variety of settings, this book develops a conceptual model. This model represents the big picture of motor performance. As new topics are introduced, they are added to the model, tying together most of the major processes and events that occur as performers produce skills. Models of this type are critical in teaching and in science because they embrace many seemingly unrelated facts and concepts, linking real-world knowledge with the concepts being discussed.

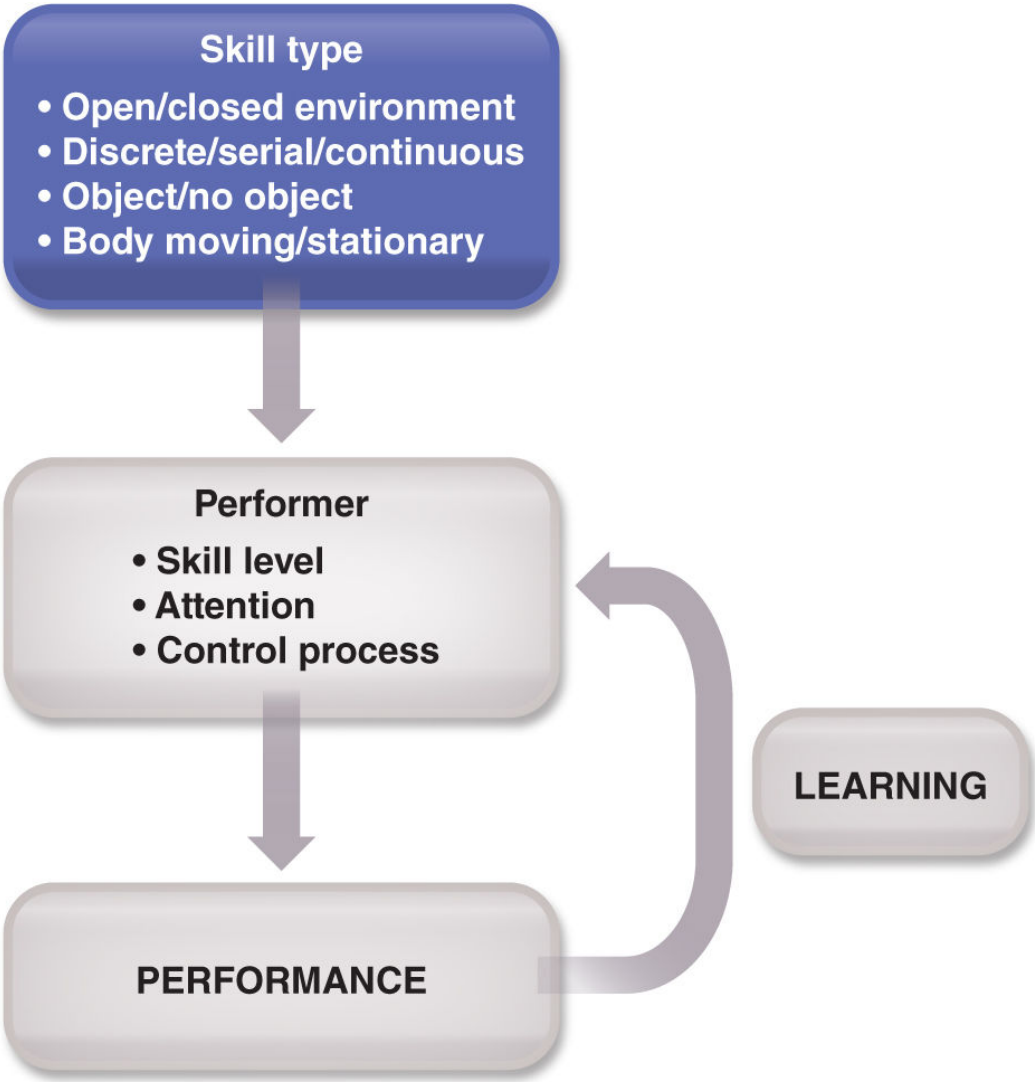

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FIGURE 1.4 An overview of the processes involved in motor learning and performance. The next part of the book is devoted to the processes that influence performance. The latter part concerns issues that influence learning.

Models can be of many types, of course, such as the plumbing-and-pump model of the human circulatory system and the variety of balls that model the structure of atoms, the solar system, and molecules in chemistry. For skills, a useful conceptualization is an information-flow model, which considers how information of various kinds is used in producing and learning a skilled action. The first portions of the text build this model, first considering how the sensory information that enters the system through the receptors is processed, transformed, and stored. To this is added how this sensory information leads to other processes associated with decision making and planning action. To the emerging conceptual model are then added features of the initiation of action as well as the activities involved while the action is unfolding, such as controlling muscular contractions and detecting and correcting errors; this is highly related to the performer's analysis of the sensations produced as a result of performing the action—processes related to feedback. In the second portion of the text, which deals with learning, the model provides an effective understanding of the processes that are, and are not, influenced by practice.

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Summary

People regard skills as an important, fascinating aspect of life. Knowledge about skills has come from a variety of scientific disciplines and can be applied to many settings, such as sport, teaching, coaching, industry, and physical therapy.

Skill is usually defined as the capability to bring about some desired end result with maximum certainty and minimum time and energy. Many different components are involved; major categories are perceptual or sensory processes, decision making, and movement output. Skills may be classified along numerous dimensions, such as open versus closed skills, and discrete, serial, and continuous skills. These classifications are important because the principles of skills and their learning often differ for different skill categories.

The text's particular organization of materials should facilitate an understanding of skills. After this introduction, the remainder of part I addresses the principles of human skilled performance and the underlying processes, focusing on how the various parts of the motor system act to produce skilled actions. Part II examines how to modify these various processes by practice and motor learning. Understanding how all of these components can operate together is facilitated by a conceptual model of human performance that is developed throughout the text.

WEB STUDY GUIDE ACTIVITIES

The student web study guide offers these activities to help you build and apply your knowledge of the concepts in this chapter.

Interactive Learning

Activity 1.1: Classify skills as discrete, serial, or continuous in nature by selecting the appropriate category for each of five examples.

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Activity 1.2: Review the types of error measured in motor learning research by matching various error measures with their definitions.

Principles-to-Application Exercise

Activity 1.3: The principles-to-application exercise for this chapter applies the concepts in this chapter to your own experience by asking you to analyze a motor skill that you have learned in the past or are currently learning. You'll describe the skill as open or closed and as discrete, serial, or continuous and identify the goals of the skill and the factors that contribute to your success when performing it.

Motor Control in Everyday Actions Narratives

Public Opinion Polls

Cutting Wood and Missing Putts

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Check Your Understanding

1. Define a skill and indicate why each of the following terms is important to that definition.
   - Environmental goal
   - Maximum certainty
   - Minimum energy costs
   - Minimum time
2. Distinguish between open and closed skills and between discrete, serial, and continuous skills. Give one example of each.
3. List and describe three elements critical to almost any skill.
4. Define a theory and describe how scientists use theories to design experiments.

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Apply Your Knowledge

1. List three motor skills that you have learned, either recently or when you were younger (e.g., swinging a baseball bat, tying your shoelaces, or playing a chord on the piano). Classify each of the skills you have listed, distinguishing between open and closed and between discrete, serial, and continuous skills. Are maximum certainty, minimum energy costs, and minimum time equally important for each of the tasks you have listed? Why or why not?

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Suggestions for Further Reading

Historical reviews of motor skills research were conducted by Irion (1966), Adams (1987), and chapter 1 in Schmidt et al. (2019). The first edition of *Motor Control and Learning* (Schmidt, 1982) contains a chapter devoted to the scientific study of motor skills. A biography of Paul Fitts was written by Pew (1994). The book by Snyder and Abernethy (1992) contains an autobiographical chapter on the early stages of the career of Franklin M. Henry. Chapter 2 of Schmidt et al. (2019) goes into much greater detail about error measures and experimental methods. Skills classification systems are reviewed in Poulton (1957), Gentile (1972, 2000), and Farrell (1975). See the references list for these additional resources.

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